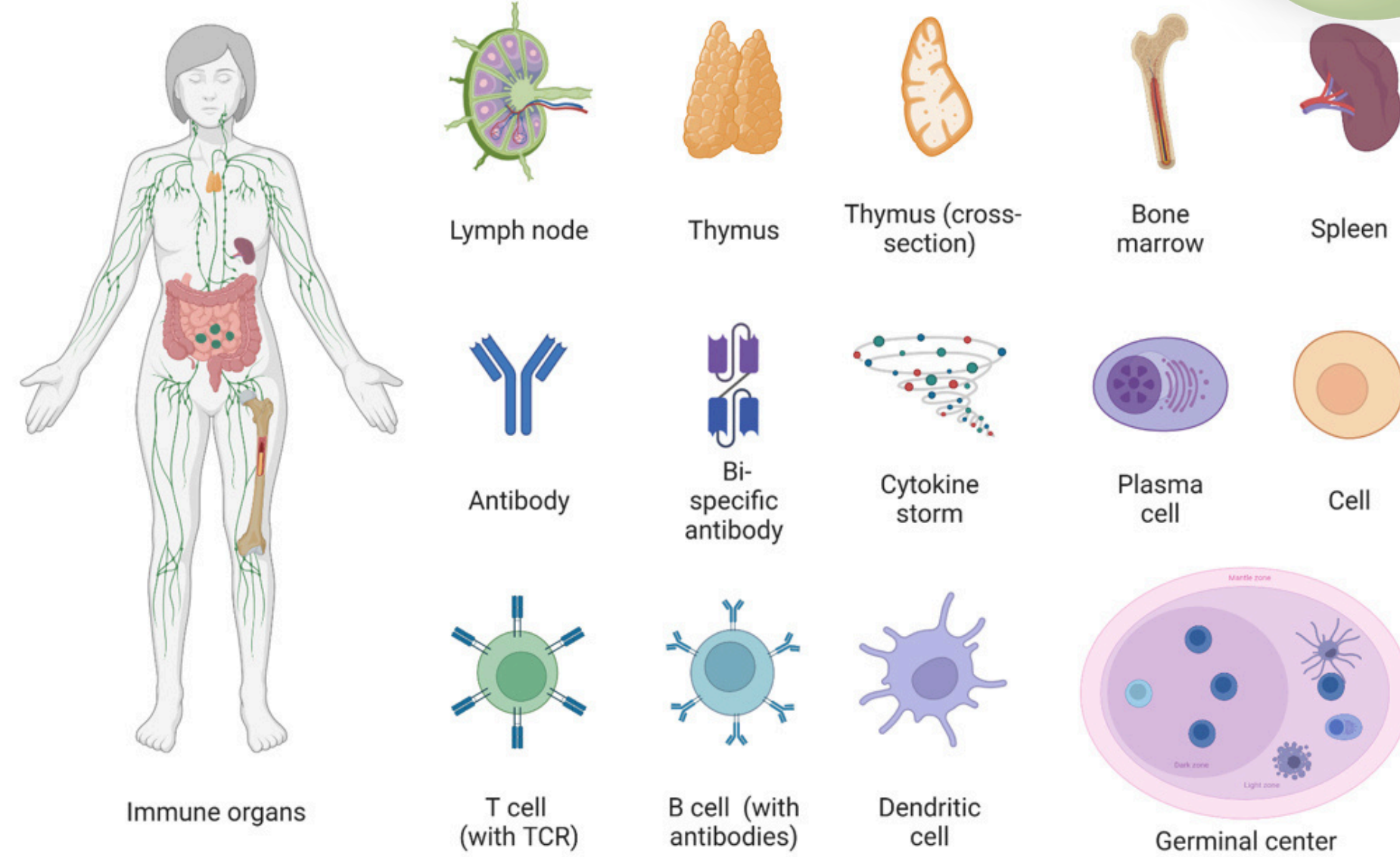


The Immune System

Gabriel, Shirley, Minh

Group 9

Outline of Immunology



First Line of Defense: Barriers

- Physical: Skin, mucous membranes, cilia (hair-like structures in airways).
- Chemical: Tears, saliva, stomach acid, sweat, and beneficial microbes.

Second Line of Defense: Innate Immunity

- Activated if pathogens breach the first line.
- Phagocytes (macrophages, neutrophils) engulf invaders.

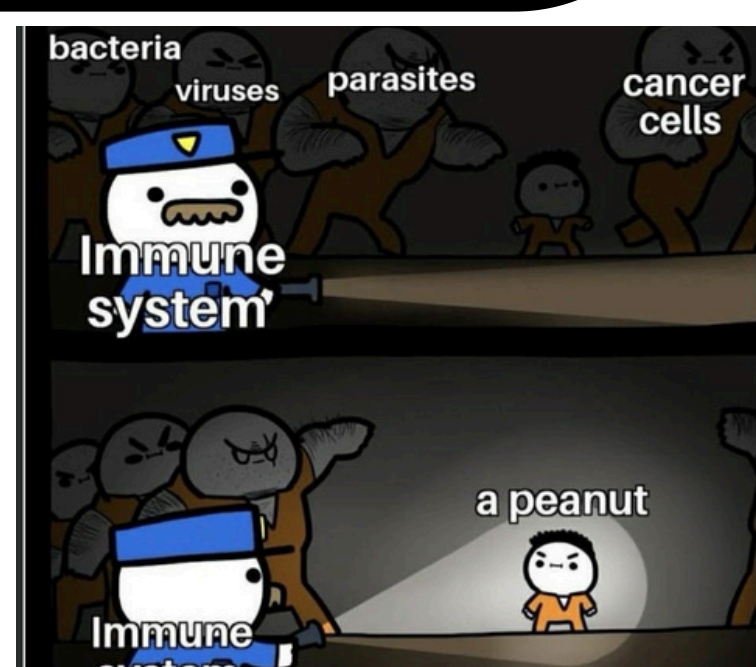
Third Line of Defense: Adaptive Immunity

- Targeted response to specific pathogens (antigens). Creates long term immunity for faster future responses.
- B Cells, T Cells: Help identify pathogens (helper T cells) and kill infected cells (cytotoxic T cells).

Fun Facts

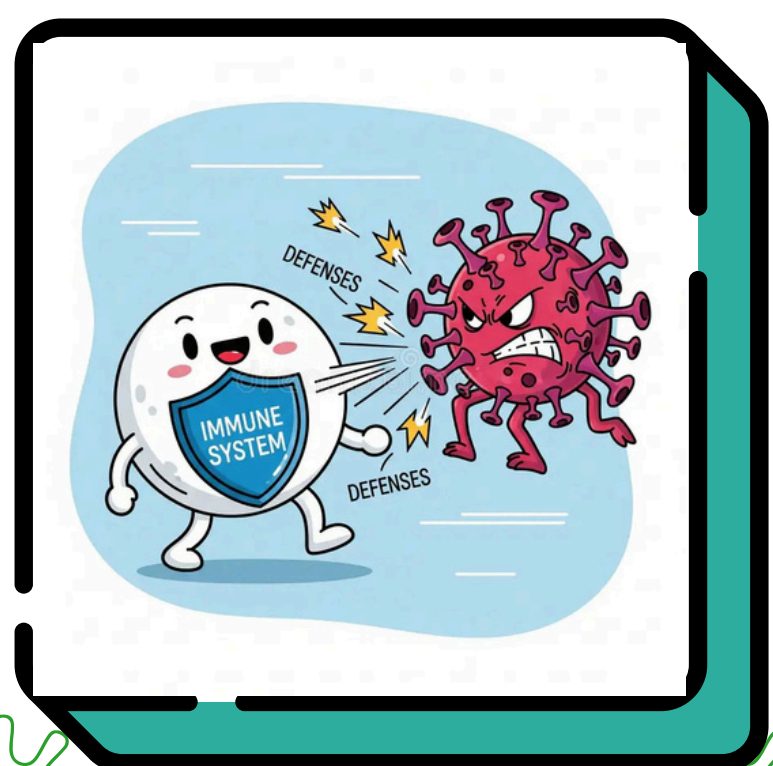
1. **Adenoids** (part of the immune system) disappear (shrink) when we grow up because **other types of defenses** will show up.

2. Sometimes, the immune system is **not completely accurate**, it can still detect **unharmful** things as very dangerous **threats**.



3. In the human body, scientists estimate that **millions to billions** of cells with **cancer-like mutations arise every day**. Thanks to the **immune system**, you are **alive**.)

Why it's important?



Protect us against abnormal cells

Defend against and remember past infections

Help you recover faster and keep your body in balance

Introducing T-cells

The **2025** Nobel Prize in Physiology or Medicine was awarded to 3 scientists for establishing the **role of Regulatory T cells (Treg cells)** in our immune system

2025 Nobel Prize winners



What we assumed before

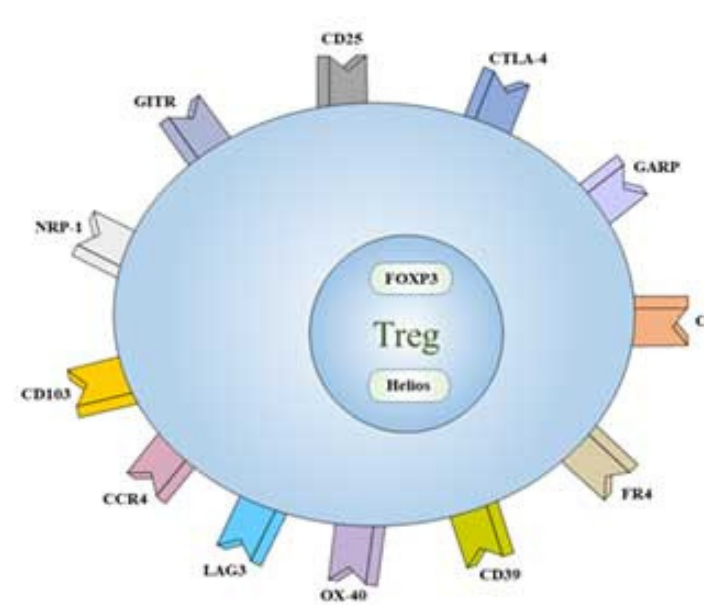
Autoimmune disease was prevented by the **DELETION** of **self reactive T cells** in the thymus

What's wrong?

Scientists realised **MANY** self reactive T cells **ESCAPE DELETION**, yet most people **DO NOT** develop autoimmune disease

What are Tregs?

A type of Tcell with the phenotype: CD4⁺ CD25⁺ FOXP3⁺



Above is a Treg cell. The things jutting out are signals so the Treg can communicate with other immune cells

Essentially,

Tregs are a form of **PERIPHERAL IMMUNE TOLERANCE**, where they act as **secondary** immune responses to aid central tolerance (**DELETION**)

What they do

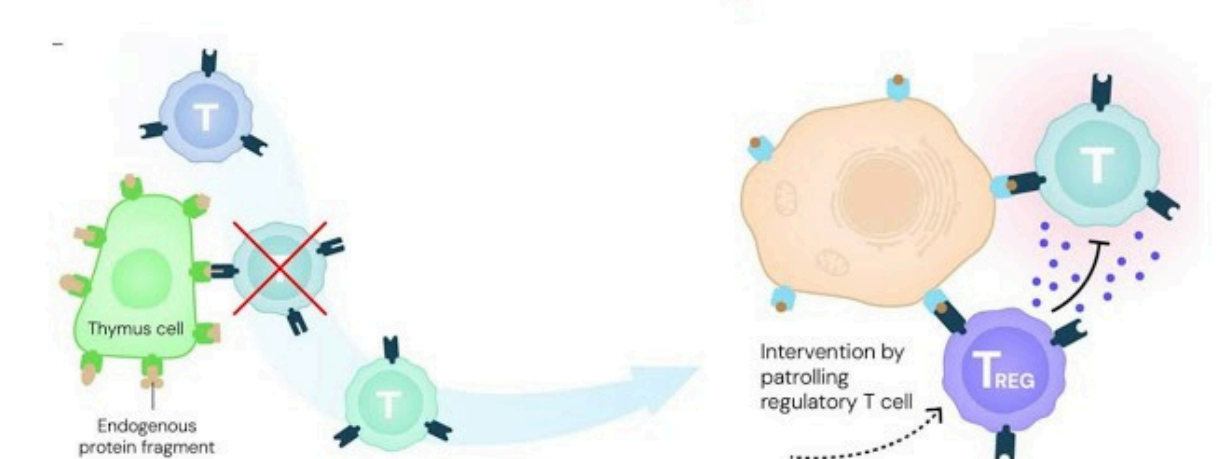
Release inhibitory cytokines to **CALM DOWN** activated T cells

Absorbs IL-2 which **starves** overactive T cells of survival signals

Reduce antigen-presenting cells' ability to activate T cells

Patrol tissues and **suppress** **SELF REACTING T CELLS**

Central tolerance vs Peripheral tolerance



Above is an illustration for the difference between central tolerance and the **NEWLY DISCOVERED** peripheral tolerance

IMPORTANCE

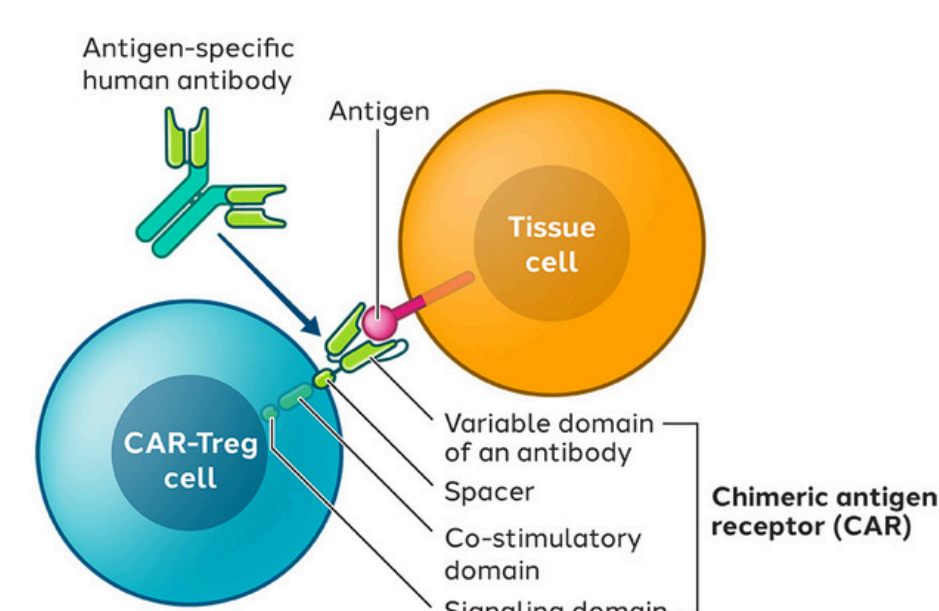
Treg **DEFICIENCY** correlates with conditions like **Type 1 diabetes**, **multiple sclerosis**, **lupus**, and **rheumatoid arthritis**.

THEREFORE, Potential therapies aim to **boost Treg number or function** to restore tolerance.

In The Future...

← **Therapeutic Engineering** like CAR-Treg therapies tailored for autoimmunity

PRECISION MEDICINE, using Tregs as **targeted delivery systems**.



Above is an illustration of how CAR-Treg therapy works

SOURCES:
Sakaguchi et al., Cell, 2008
Josefowicz et al., Nature Immunology, 2012